Chapter 15 – Natural Language Processing (NLP), RNNs & LSTMs

> *Estimated Duration:* *10–14 Days*
>
> *Difficulty:*
>
> *Priority:*
>
> *Goal:* *Understand how computers process human language, learn sequence models, understand why RNNs and LSTMs were created, and build text classification systems using PyTorch.*

Introduction

Until now, we have worked with

- Numbers
- Tables
- Images

Now we move to the most challenging type of data:

Human Language.

Language is very different from images.

Consider these two sentences:

```
I love AI.
```

```
AI loves me.
```

Both contain the same words.

But they mean completely different things.

Why?

Because **word order matters.**

Traditional Neural Networks don't naturally understand sequences.

This led to the development of

- Recurrent Neural Networks (RNNs)
- Long Short-Term Memory Networks (LSTMs)

What is NLP?

NLP stands for

Natural Language Processing

It is the field of AI that enables computers to understand, generate, and interact using human language.

Examples

- ChatGPT
- Google Translate
- Siri
- Alexa
- Grammarly
- Spam Detection
- Sentiment Analysis
- Search Engines

Why is Language Difficult?

Computers understand numbers.

Humans communicate using words.

Before a computer can understand text,

we must convert language into numbers.

Example

```
Hello


125
```

or

```
Hello


[0.12, -0.44, 0.81, ...]
```

This conversion is the foundation of NLP.

Learning Objectives

By the end of this chapter you should be able to:

- Explain NLP.
- Understand sequence data.
- Explain RNNs.
- Understand the Vanishing Gradient problem.
- Explain LSTMs.
- Understand GRUs.
- Perform text preprocessing.
- Convert text into vectors.
- Build text classification models.

NLP Roadmap

Text

Cleaning

Tokenization

Vocabulary

Encoding

Word Embeddings

RNN

LSTM

GRU

Text Classification

Projects

Module 1 – Text Preprocessing

Estimated Time

1 Day

Real-world text is messy.

Example

```
I LOVE AI!!!!
```

Cleaning steps

- Lowercasing
- Remove punctuation
- Remove HTML
- Remove URLs
- Remove special characters
- Remove emojis (task-dependent)
- Normalize whitespace

Good preprocessing improves model quality.

Module 2 – Tokenization

Estimated Time

Half Day

Tokenization means

breaking text into smaller units.

Example

```
I love AI
```

```
["I", "love", "AI"]
```

Tokens become the basic units of language models.

Module 3 – Stop Words

Estimated Time

Half Day

Some words occur very frequently.

Examples

- the
- is
- a
- an
- of

Sometimes they add little value.

Sometimes they are important.

Modern NLP often keeps them,

but you should understand the concept.

Module 4 – Stemming & Lemmatization

Estimated Time

Half Day

Different words may have the same root.

Examples

> Playing
>
> Played
>
> Plays

□

> Play

These techniques reduce vocabulary size.

Module 5 – Text Representation

Estimated Time

1 Day

Computers cannot understand words directly.

We convert text into numbers.

Learn

- Bag of Words
- TF-IDF
- Word Embeddings (Introduction)

Understand the strengths and weaknesses of each.

Module 6 – Word Embeddings

Estimated Time

1 Day

Instead of assigning one number,

represent each word as a vector.

Example

```
King


[0.2, -0.6, 0.9, ...]
```

Words with similar meanings have similar vectors.

Examples

King ≈ Queen

Paris ≈ France

Delhi ≈ India

This idea revolutionized NLP.

Module 7 – Why RNNs?

Estimated Time

1 Day

Traditional Neural Networks

process inputs independently.

Language is sequential.

The meaning of a word depends on previous words.

RNNs introduce

memory.

Instead of forgetting the previous word,

they pass information forward.

Mental Model

Imagine reading a sentence.

You don't forget the previous word after reading the next one.

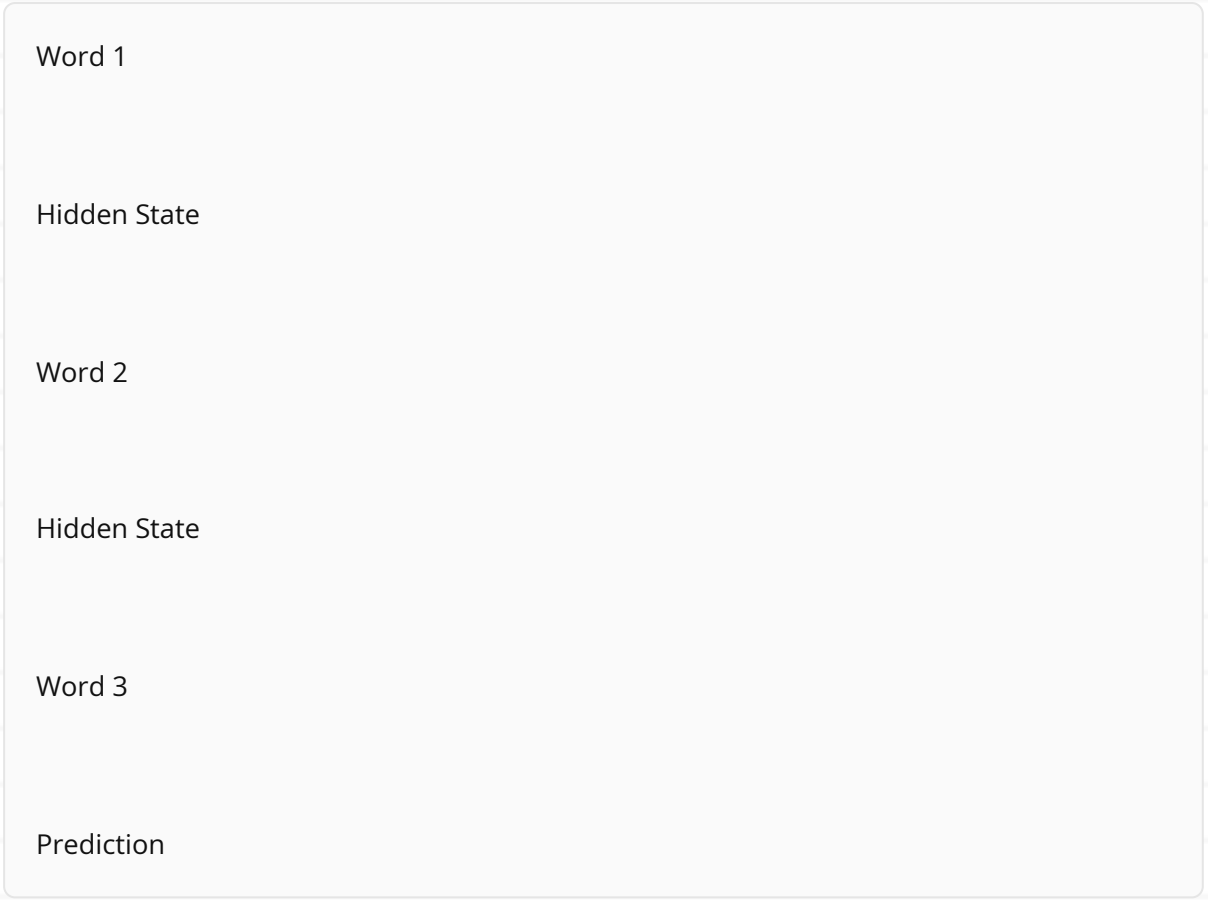
RNNs try to imitate this behavior.

Module 8 – Recurrent Neural Networks (RNNs)

Estimated Time

1 Day

Workflow



The hidden state carries information from earlier words.

Module 9 – Vanishing Gradient Problem

Estimated Time

Half Day

Problem

As sequences become longer,

RNNs gradually forget earlier information.

Example

Sentence

The movie released in 1998 was surprisingly good because...

The model may forget the beginning before reaching the end.

This limitation motivated LSTMs.

Module 10 – Long Short-Term Memory (LSTM)

Estimated Time

2 Days

LSTMs improve RNNs by introducing

memory cells

and

gates.

Types of Gates

- Forget Gate
- Input Gate
- Output Gate

These gates decide

- What to remember.
- What to forget.
- What to output.

LSTMs handle long sequences much better than standard RNNs.

Module 11 – GRU (Introduction)

Estimated Time

Half Day

GRU

(Gated Recurrent Unit)

A simpler alternative to LSTM.

Advantages

- Faster training.

- Fewer parameters.
- Often similar performance.

Module 12 – Sequence Classification

Estimated Time

1 Day

Workflow

```
Text

Tokenization

Encoding

Embedding

LSTM

Prediction
```

Applications

- Spam Detection
- Sentiment Analysis
- News Classification

Module 13 – Evaluation

Estimated Time

Half Day

Metrics

Accuracy

Precision

Recall

F1 Score

Confusion Matrix

Always inspect

misclassified examples.

Module 14 – PyTorch NLP Pipeline

Estimated Time

1 Day

Complete workflow

Dataset

Tokenizer

Vocabulary

Encoding

Padding

Embedding

LSTM

Loss

Optimizer

Training

Evaluation

Mini Exercises

Practice

- Spam Detection.
- Movie Reviews.

- News Classification.
- Twitter Sentiment.
- SMS Spam.
- Fake News Detection (basic).

Projects

Project 1 – SMS Spam Detection

Binary Classification

Spam

Not Spam

Project 2 – Movie Review Sentiment Analysis

Predict

Positive

Negative

Project 3 – News Category Classification

Categories

Sports

Politics

Business

Technology

Entertainment

Project 4 – Toxic Comment Detection

Identify harmful online comments.

Project 5 – Customer Review Analysis

Predict customer satisfaction from product reviews.

Interview Corner

Common Questions

- What is NLP?
- Why can't normal Neural Networks handle language well?
- What is Tokenization?
- Difference between Bag of Words and TF-IDF?
- What are Word Embeddings?
- What is an RNN?
- Why were LSTMs invented?
- Difference between RNN, LSTM, and GRU?

Industry Insight

RNNs and LSTMs were groundbreaking.

But today,

most modern NLP systems,

including ChatGPT,

no longer use them as the primary architecture.

Instead,

they use

Transformers.

However,

understanding RNNs and LSTMs helps explain the problems Transformers solved.

Common Mistakes

Ignoring preprocessing.

Treating words independently.

Forgetting sequence length.

Using Bag of Words for every NLP task.

Ignoring class imbalance.

Best Practices

 Clean text carefully.

 Choose preprocessing based on the task.

 Start with simple baselines.

 Compare TF-IDF and embeddings.

 Analyze model errors.

 Understand why sequence order matters.

Skills Achieved

After this chapter, you should understand:

NLP fundamentals.

Text preprocessing.

Tokenization.

Word embeddings.

RNNs.

LSTMs.

GRUs.

Sequence classification.

You now understand the historical foundation of modern NLP.

Completion Checklist

Before moving forward, make sure you can:

- [ ] Explain why NLP is challenging.
- [ ] Perform text preprocessing.
- [ ] Tokenize text.
- [ ] Explain Bag of Words, TF-IDF, and Embeddings.
- [ ] Explain why RNNs were created.
- [ ] Explain the Vanishing Gradient problem.
- [ ] Build an LSTM model in PyTorch.
- [ ] Complete at least three NLP projects.

Chapter Summary

Congratulations.

You now understand how AI processed language before the Transformer era.

RNNs and LSTMs introduced the idea of remembering previous words,

but they still had major limitations:

- Slow training.
- Difficulty processing long sequences.
- Limited parallelization.
- Loss of long-range context.

In 2017,

a research paper changed NLP forever.

Its title was:

> *Attention Is All You Need*

That paper introduced the **Transformer** architecture,

which powers modern AI systems such as ChatGPT, Claude, Gemini, Llama, and many others.

Everything you've learned so far has been preparing you for this moment.

What's Next?

Chapter 16 – Transformers & Large Language Models (LLMs)

You'll learn:

- Why Transformers replaced RNNs.
- Self-Attention.
- Multi-Head Attention.
- Positional Encoding.
- Encoder vs Decoder.
- The Transformer architecture.
- BERT vs GPT.
- Tokenizers used in LLMs.
- Pretraining vs Fine-tuning.
- How ChatGPT actually works at a high level.

> ***Golden Rule:*** *Transformers didn't just improve NLP—they fundamentally changed AI. Understanding Transformers is one of the most valuable skills in modern Artificial Intelligence.*